

UNIT 10

WHAT IS WEB 3.0? A BRIEF INTRODUCTION AND IT'S BENEFITS

If you ask someone, they'll tell you Web 2.0 as we know it is probably on its way out the door. For many, Web 2.0 is characterized mainly by the ability of users to share information quickly with others, which has been developed into the phenomenon that we call social media.

From Twitter to Facebook to YouTube and to all sorts of other kinds of communities, Web 2.0 is all about sharing and seeing. Now if you recall or were around during what is now known as Web 1.0, information was put up on a website and that was it—the best way of sharing it was privately through e-mails and such.

There was little to no communication and if you wanted information, you had to go to the source for it. Can you imagine such a harsh internet? Now with Web 2.0 on its way out, the obvious question is, what in the world is Web 3.0 going to be?

What is Web 3.0?

If you're anything like me, it's hard to imagine how the internet is going to top sites like Twitter and Facebook. But it's bound to happen and when you research Web 3.0, you find out it is going to be synonymous with the user's interaction with the web.

In Web 2.0 we focused on the users' interaction with others, now we are going to focus more on the users themselves, which is always a plus. But how is this going to happen?

Web 3.0 is being referred to by experts as the semantic web; semantic meaning data driven. The data will come from the user and the web will essentially adjust to meet the needs of the user. For example, if you do a lot of searching for 'design blogs', you'll receive more advertisements related to design.

Also, when you search for other things, for example, 'computers', the web will keep in mind that you often search for design and may pull up search queries that combine 'design' and 'computers'.

Benefits of Web 3.0

A huge benefit of Web 3.0 is the move towards being able to access data from anywhere. This is mainly being driven by the heavy usage of smart phones and cloud applications.

The idea here is to make sure that the user can access as much data as possible from anywhere, not just their home. Technology is trying to expand this idea in ways that

allow TVs to pick up on user data and allowing smart phones to access data on your computer.

For designers, who typically forget their jump drives, this is an amazing and useful advancement!

Web 3.0's Effect on Design

So now that you have an idea of what Web 3.0 is and what it's going to be, we have to ask the most important question for us: what does that mean for design? Web 2.0 design was based around drawing attention and persuading your audience, because after all, Web 2.0 made a huge deal about being able to purchase things online.

Web 2.0 wanted to generate excitement and get people to make a purchase and understand what they were doing. You want to make a purchase? Sure, then click this button. You want to join the mailing list? Great, then there's no question about clicking this button. That is the basis of Web 2.0 design.

Other elements were added to make things more fun and give a bit of style. The usage of linear gradients in Web 2.0 is almost necessary.

Whatever color combination you desire, linear gradients are typically present from your background to your buttons. Other trends surfaced like various badges, rounded corners and the necessary usage of icons. But again, the question remains, what can we expect for Web 3.0?

Web 2.0 Design vs. Web 3.0 Design

In Web 2.0 we had to create a design that was great for the web. I think in Web 3.0, we will firstly have to create a design that is going to be good not just for the web and the web browser, but for all sorts of media. With the growth in the usage of smart phones and tablets, people want more usage out of their items and to be able to access more things as best as possible.

Design will have to be able to translate in great quality across all sorts of technologies. Now while you can create two different websites (one for the web and one for mobile devices), designers and developers will have to kill two birds with one stone, by creating one website that will look good in both environments.

Also, for Web 3.0, designers will continue to focus on making things simpler. The truth is the designer has the absolute power to persuade viewers on where to look first and second and so forth and so on. By doing this the designer creates a hierarchy of importance, that should not be muddled by useless design.

Designers will continue to design so that content remains king by putting much focus on it and taking focus off non-content things such as logos and navigation bars.

Web 3.0 Design Trends

Using these types of techniques plays into the increasing popularity of the minimalist design technique, where the focus is not necessarily making something as simple as possible, but making it as simplistic as possible.

Creating a site with non-flashy web elements makes the user HAVE to focus on the content of the site. Of course, designers desire to design and will 'fancify' some things, but in Web 3.0, that isn't the main focus. The focus is to draw the viewers' eye to the content or other important information on the page.

Many of the design trends used in Web 2.0 will only change by way of design, but not really the usage. The change in Web 2.0 to Web 3.0 is about how the internet is used, not really how it's seen (with the exception of mobile devices and such).

I believe designs will continue to get more and more minimalistic while maintaining a certain sense of beauty, but of course we will continue to use buttons and rounded corners and gradients. The design of Web 3.0 will be based on the way designers decide to design it and what becomes popular.

Exercises

1. Answer the following questions.

- a) What is Web 2.0 mainly characterized by?
- b) What are the differences between Web 2.0 and Web 1.0?
- c) What are the differences between Web 2.0 and Web 3.0?
- d) Give some benefits of Web 3.0.
- e) What were the purpose and features of design in Web 2.0?
- f) What components will the design of Web 3.0 (have to) include, according to the text?
- g) Explain the purpose of “minimalist design technique” in the design of Web 3.0.
- h) What the “fancifying of things” imply in terms of web design?
- i) What is in the focus of a Web 3.0 site design?
- j) In which way will many of the design trends used in Web 2.0 change and why?
- k) According to the author of the text, what is the future of (web) designs and will the design of Web 3.0 be based on?

2. Explain/describe the following terms and expressions, on the basis of their use in the texts. Use a dictionary or check online if necessary.

social media	
obvious	
synonymous	
semantic web	
data driven	
semantics	
pick up on	
jump drive	
draw attention	
linear gradient	
surface	
translate across	
kill two birds with one stone	
persuade	
muddled	
simplistic	
minimalistic	

3. Connect the sentence parts in accordance with the text and its order to get correct sentences.

Web 2.0 is characterized mainly by the ability of users to share information quickly with others,	and this is mainly being driven by the heavy usage of smart phones and cloud applications.
In Web 1.0, information was put up on a website and that was it –.	not really how it's seen (with the exception of mobile devices and such).
Web 3.0 is being referred to by experts as the semantic web; semantic meaning data driven,	not just for the web and the web browser, but for all sorts of media.
A huge benefit of Web 3.0 is the move towards being able to access data from anywhere	people want more usage out of their items and to be able to access more things as best as possible.
Web 2.0 wanted to generate excitement and get people to make a purchase	which has been developed into the phenomenon that we call social media.
In Web 3.0, we will have to create design that is going to be good	the best way of sharing it was privately through e-mails and such
With the growth in the usage of smart phones and tablets,	the focus is to draw the viewers' eye to the content or other important information on the page.
The change in Web 2.0 to Web 3.0 is about how the internet is used,	and understand what they were doing.
Designers desire to design and will 'fancify' some things, but in Web 3.0, that isn't the main focus:	where the data will come from the user and the web will essentially adjust to meet the needs of the user.

4. Complete the text with the given words and expressions.

| paradigm-shifting | information | scientists | social media | marketing | internet
 | concept | streaming | advertise | visitors | upload | evolution | websites |
 minimalistic | e-commerce |

What is Web 3.0? The Evolution of the Internet

You have probably heard the term "web 3.0" floating around the _____. Simply put, web 3.0 is the new phase of the internet's _____. The changes that web 3.0 is bringing to the internet is going to take it to a whole new level. Computer _____ and Internet experts believe that these changes are going to make the internet smarter and our lives easier. So, to understand these _____ changes, let's first look at the evolution of the internet as we know it.

As we are going to see soon, the internet has evolved along with our needs and uses.

Web 1.0

Believe it or not, there used to be a version of the internet that existed before _____ and video _____.! This was a time before Google in the mid-to-late '90s. The internet used to be dominated by AltaVista and Netscape. Back then the internet existed to only _____ their brick-and-mortar companies. These websites were “read-only web,” meaning you were only allowed to search for _____ and read it.

Most _____ websites are still web 1.0 in nature since the _____ behind them is simple. Present products to the customers and take money from the ones who are interested. These _____ are usually very smooth and fast, however, the level of user interaction is _____.

Web 2.0

The next iteration of the internet was called the “web 2.0” or the “read-write” web. Now, users were not just idle _____, they could create their own content and _____ it to a website. Starting roughly around 2003 when the term was coined by Dale Dougherty, Vice President at O'Reilly Media, web 2.0 has taken over the world by storm. In just over a decade, it has completely redefined _____ and business operations.

| e-shopping | intelligent | institution | share | Influencers | learning |
information/data | user-accessible | decisions | algorithm | nutshell | review |
online applications | recommendation | content |

Instagram _____ can make or break a brand by posting one single photo, Yelp reviewers can destroy a restaurant by one single negative _____. Audience reviews are critical when it comes to buying _____. According to a survey, 90 percent of customers reading online reviews before purchasing and 88 percent of them trusting them as much as a personal _____.

Web 2.0's main aim was to make the internet more democratic and make it as _____ as possible.

Web 3.0

Every time you buy something on an _____ website, the website's _____ will look at the other items that people who have purchased your product went on buy and then recommends that to you. So, think about what is going on here. The website is _____ from other users what your preferred choices can be and then use it to recommend to you what you may like. In essence, the website itself is learning and becoming more _____.

That, in a _____, is the very philosophy behind web 3.0. Web 1.0 was primarily driven by _____ that came from the business or the

_____ for its customers. Web 2.0 took things a little bit further by allowing users to upload and _____ their content on the website itself. Web 3.0 allows _____ and websites to receive information that's on the Web and give new _____ to the users.

UNIT 11

WHAT IS THE SEMANTIC WEB?

The Semantic Web is the knowledge graph formed by combining connected, Linked Data with intelligent content to facilitate machine understanding and processing of content, metadata, and other information objects at scale.

The Semantic Web leads to smarter, more effortless customer experiences by giving content the ability to understand and present itself in the most useful forms matched to a customer's need. Semantic standards unlock a crucial evolution of the web towards intelligence that allows the content we post online to be presented in a way that can be understood, connected, and remixed by machines.

Semantic Web content structures form an essential basis for a reliable graph, or map of knowledge, necessary for true artificial intelligence (AI) beyond basic Natural Language Processing (NLP) and Natural Language Understanding (NLU). AI will always remain niche applications built against a limited corpus of content until structure and semantic standards exist across content sets. Adopting Semantic Web approaches to content gets publishers closer to globally-machinable sets of content.

Engineering the Semantic Web

Content engineers are creating a more powerful and agile web of content and data by first parsing and structuring the discrete elements of content that constitute websites, such as people, events, ideas, concepts, products. These elements are then assigned a "label" describing its meaning in a standardized language. When such machine-readable descriptions are present, they can be linked to build a more robust web of data where computers can find, read, and even reason about a unit of content.

We can see the application of semantic data in various places throughout the web, such as those in certain search experiences. Because of this rich, new layer of information, search engines and other bots are able to provide the most relevant content directly to the user, edited to the most important snippets that save humans time and effort.

The Semantic Web not only improves traditional search, but it is facilitating more seamless, intelligent, and integrated customer experience journeys as well. For example, with semantically connected and described data, a digital assistant could send users local live music recommendations in their area. This could be possible by gathering and connecting disparate data published on the web, like information that nearby venues post online, and matching it to the data about the type of music a user has chosen to share on their online playlists.

The applications of the Semantic Web are endless, but we cannot take advantage of these possibilities until we have a truly intelligent web of global knowledge. We must make our content “semantic”, or annotated with meaningful metadata and relationships, to transform dull and dormant fixed-text into live and electric linked concepts. This transformation is making the web much more dynamic, allowing not only content, but also data, to travel freely and seamlessly.

The Workings of the Semantic Web

To understand the principle behind how the Semantic Web evolved, imagine a jukebox. This classic machine plays the song a patron selects through the push of buttons. As the jukebox contains a limited number of recordings that must be selected manually, the web before semantic technologies were introduced worked in much the same way and had many of the same limitations. Users had to manually pull requests from limited resources: web pages, directories, documents residing on different servers, etc. Machines could not find, read, or let alone use this content.

Through the decades, on-demand music technology has evolved from jukeboxes to give us intelligent digital platforms like Spotify or Pandora. These “Semantic Music” services let us discover the right content out of millions of songs without listening to all of them, build human- and machine-curated collections of music, and perhaps most excitingly — “magically” discover new songs that match our interests.

How do Spotify and Pandora do this? Or how do their cousins Netflix, YouTube, AirBNB and Amazon manifest their forms of “customer experience magic”? You guessed it! Through the use of semantically rich Linked Data and content.

The web has been evolving towards semantics since it was a glimmer in Tim Berners-Lee’s vision. Instead of humans manually searching from a limited list of links, now algorithms navigate a vast amount of increasingly structured content sets available to answer specifically or act upon a certain query. Data, enriched with semantics, structure, and meaningful, machine-interpretable links allows computers to find and manipulate information on our behalf with greater precision. This results in better experiences of content discovery and search, and broader opportunities for the seamless share, recombination, analysis, and reuse of data with less human-manual-human interaction in the loop. Semantic experiences make it easier for humans to perform creatively and encounter frictionless content experiences, while a machine does the finding, sifting, sorting, combining, organizing, and presenting of real answers.

A Brief Introduction to the Technical Standards of the Semantic Web

The Semantic Web provides a lingua franca for representing and working with data. The technological framework to bring the semantic vision to life is built on standards, developed, agreed upon, and published by the World Wide Web

Consortium (W3C). These standards aim to provide a uniform and interoperable way of representing and managing data and content so that it can be easily shared and reused across applications, enterprises and individuals.

The technology stack that supports the Semantic Web is designed to enable computers, software systems, and people to work together in a network. It consists of a wide array of technologies, the most important of which are: RDF, SPARQL and OWL.

RDF

RDF stands for Resource Description Framework and is a method for publishing and linking data. It is built around the following Web standards: XML and URL (URI) and uses the model of the “triple” — three elements (namely a subject, a predicate and an object) bound together, with the relationships between them formally described. Triples, when there are enough of them, form a rich graph of knowledge.

SPARQL

SPARQL stands for Protocol and RDF Query Language and is the way data stored in RDF format gets queried, retrieved and manipulated on the Semantic Web. It is SPARQL that allows us to navigate diverse databases and discover relationships between data. Those familiar with SQL will notice the differences in capabilities between the query languages.

OWL

OWL stands for the W3C Web Ontology Language. OWL is a logic-based language, designed to be interpreted by machines so that they can automatically process and reason through the information it describes.

How Do We Get Started With the Semantic Web?

The Semantic Web takes a village. We need to graph our own content sets and connect them to globally related content sets. Start locally. Start at home, in our own enterprises. Then reach out to connect to the content within related content ecosystems of customers, partners, vendors, and even competitors.

Those of us in the content industry, or anyone involved in content publishing inside an enterprise, can seamlessly build this semantic intelligence into our content sets, but not overnight. A nimble structure for knowledge takes discipline and the application of patterns over time. It is through the practice of content engineering, in connection with smart content strategy, that allows content ecosystems to be designed for these rich semantic connections.

Why Should We Invest in Making Our Content Semantic?

The spread of the Semantic Web and the technologies it brings to the table put the analytical powers of machines to work in the domains of content production, management, learning, support, media, ecommerce, scientific research, knowledge management, and publishing in general. Any place in which we express knowledge will become semantic. When it comes to the applications of intelligent content, semantic search and smart devices, the emerging semantic web of content and data is a massive opportunity to tap into. Careers, companies, and global innovation leaders will continue to be born on the Semantic Web.

One could say that the Semantic Web is the technological backbone of the evolving content order, bringing together omnichannel content with semantics, structure, and shared standards.

Using Semantic Web Technologies, publishers can:

- Build smart digital content infrastructures
- Connect content silos across a huge organization
- Leverage metadata to provide richer experiences
- Curate and reuse content more efficiently
- Connect internal and external content sets
- Build towards real augmented and artificial intelligence
- Power-up authoring experiences and workflow processes

In order to engineer in such a way that we are planning for the changing content ecosystems, we need to understand the significance of semantic data connections and begin to incrementally embed semantic metadata and relationships into every piece of content we design.

Though the formation of the Semantic Web is a challenging and constantly evolving construct, one thing remains constant: it is steadily leading the communication between people, software agents and devices towards intelligent content infrastructures and a smarter collaborative future for humans.

Exercises

1. Answer the following questions.

- a) What is the Semantic Web?
- b) What does the Semantic Web lead to?
- c) How do the semantic standards unlock a crucial evolution of the web?
- d) What do the Semantic Web content structures form?
- e) What are the niche applications for Semantic Web?
- f) What does the adopting of Semantic Web approaches to content get publishers closer to?
- g) How are the content engineers creating a more powerful and agile web of content and data?
- h) Give some places on the web in which the application of semantic data can be seen?
- i) How does the Semantic Web improve the customer experience on the Internet? Give an example.
- j) What is the condition necessary to take full advantage of the possibilities and applications of the Semantic Web and how should it be met?
- k) Explain the comparison of the Semantic Web with a jukebox.
- l) How has the on-demand music technology evolved from jukeboxes through the decades?
- m) Describe the way in which the web has been evolving towards semantics, through the improvement of algorithms, data, links and experiences.
- n) What is the technological framework of the Semantic Web built on?
- o) What do the Semantic Web standards aim to?
- p) What is the purpose of the technology stack that supports the Semantic Web and what does it consist of?
- q) What do you know about RDF?
- r) What do you know about SPARQL?
- s) What do you know about OWL?
- t) How should we start with the Semantic Web?
- u) Why should the investments be made in making the web content semantic?
- v) Why can the Semantic Web be called technological backbone?
- w) Name at least 4 things that publishers can do by using Semantic Web technologies.
- x) What should be done in order to engineer the Semantic Web in a way planned for the changing content ecosystems?
- y) What is the thing that remains constant, relating to the Semantic Web?

2. Explain/describe the following terms and expressions, on the basis of their use in the texts. Use a dictionary or check online if necessary.

knowledge graph	
metadata	
information objects	
at scale	
parsing	
discrete	
snippets	
seamless	
jukebox	
loop	
lingua franca	
semantic	
interoperable	
vendor	
nimble	
technological backbone	
World Wide Web Consortium	
leverage	
omnichannel content	
power-up	
incrementally	
niche applications	
content silo	
software agents	

3. Write the descriptions of the modified nouns from the table.

Semantic Web content	structures	
true artificial	intelligence	
Natural Language	Processing	
Natural Language	Understanding	
Semantic Web	approaches	
machine-readable	descriptions	
seamless, intelligent, and integrated customer experience	journeys	
semantically connected and described	data	
local live music	recommendations	

on-demand music	technology	
intelligent digital	platforms	
machine-curated	collections	
increasingly structured content	sets	
machine-interpretable	links	
human-manual-human	interaction	
frictionless content	experiences	
Resource Description	Framework	
smart digital content	infrastructures	
internal and external content	sets	
semantic data	connections	

4. Rearrange the sentence parts to get correct sentences.

Semantic standards unlock a crucial evolution	the content we post online	connected, and remixed by machines	of the web towards intelligence that allows	to be presented in a way that can be understood,
Content engineers are creating a more powerful and agile	the discrete elements of content that constitute	a “label” to these elements describing its meaning in a standardized language	websites, such as people, events, ideas, concepts, products and then assigning	web of content and data by first parsing and structuring
The Semantic Web not only improves	more seamless, intelligent,	experience journeys as well	and integrated customer	traditional search, but it is facilitating
Instead of humans manually searching	navigate a vast amount of increasingly	specifically or act upon a certain query	from a limited list of links, now algorithms	structured content sets available to answer
Semantic experiences make it easier	while a machine does the finding, sifting, sorting, combining,	and encounter frictionless content experiences,	organizing, and presenting of real answers	for humans to perform creatively

The Semantic Web standards aim	of representing and managing data and content	applications, enterprises and individuals	to provide a uniform and interoperable way	so that it can be easily shared and reused across
The spread of the Semantic Web and the technologies	of machines to work in the domains of	scientific research, knowledge management, and publishing in general	content production, management, learning, support, media, ecommerce,	it brings to the table put the analytical powers
In order to engineer in such a way that	we need to understand the significance of semantic data connections	we are planning for the changing content ecosystems,	and relationships into every piece of content we design	and begin to incrementally embed semantic metadata
Though the formation of the Semantic Web is a challenging	remains constant: it is steadily leading the	intelligent content infrastructures and a smarter collaborative future for humans.	and constantly evolving construct, one thing	communication between people, software agents and devices towards

5. Complete the text with the given words and expressions.

| embedded semantics | computers | markup | Ontology | data web | metadata | Semantic | application | sources | standards | mechanisms | domains | protocols | information science | machine-readable |

The Semantic Web is an extension of the World Wide Web through _____ set by the World Wide Web Consortium (W3C). The goal of the Semantic Web is to make Internet data _____. To enable the encoding of semantics with the data, technologies such as Resource Description Framework (RDF) and Web _____ Language (OWL) are used. These technologies are used to formally represent _____. For example, ontology can describe concepts, relationships between entities, and categories of things. These _____ offer significant advantages such as reasoning over data and operating with heterogeneous data _____.

These standards promote common data formats and exchange _____ on the Web, fundamentally the RDF. According to the W3C, “The Semantic Web provides a common framework that allows data to be shared and reused across_____, enterprise, and community boundaries.” The Semantic Web is therefore regarded as an integrator across different content and information applications and systems.

The term was coined by Tim Berners-Lee for a web of data (or_____) that can be processed by machines —that is, one in which much of the meaning is machine-readable. While its critics have questioned its feasibility, proponents argue that applications in library and_____, industry, biology and human sciences research have already proven the validity of the original concept.

Berners-Lee originally expressed his vision of the Semantic Web in 1999 as follows: I have a dream for the Web in which _____ become capable of analyzing all the data on the Web – the content, links, and transactions between people and computers. A “Semantic Web”, which makes this possible, has yet to emerge, but when it does, the day-to-day_____ of trade, bureaucracy and our daily lives will be handled by machines talking to machines. The “intelligent agents” people have touted for ages will finally materialize.

The 2001 Scientific American article by Berners-Lee, Hendler, and Lassila described an expected evolution of the existing Web to a_____ Web. In 2006, Berners-Lee and colleagues stated that: “This simple idea...remains largely unrealized”. In 2013, more than four million Web _____ contained Semantic Web_____.